

start-ups that the country has never witnessed before.

“A lot of big companies are coming to India for electronics manufacturing. These companies will need support from local players in a lot of things, and this is where the MSMEs and the electronics start-ups can come in. Mark my words, this decade will be the decade of electronics start-ups in the country,” explains Nitin.

The government of India, during the last twelve months, has announced PLI for manufacture of smartphones, air-conditioners, IT hardware products, batteries, and a lot of other products concerning the electronics sector. Though these schemes offer hundreds and thousands of millions of rupees as incentive to global and local companies, the clause that requires production to be *desi* (indigenous) does spell some big opportunities for MSMEs and start-ups working in the electronics domain.

Where are we today?

The total market of electronics in India in 2019, as per a report by Forst & Sullivan, was approximately \$160 billion. The country, as per the report, met the demand for electronics in the same year by importing 50 percent and producing 50 percent locally. While that means that around \$80 billion worth of electronics manufacturing took place in the country, we still imported \$80 billion worth of electronics that year. But there is a catch to the \$80 billion worth of electronics produced in the country.

“The biggest challenge for the electronics sector in India is the low value addition manufacturing that we are doing in the country. This is resulting in the overall valuation of electronics produced in India to less than 17 percent. That simply means that the \$80 billion worth of electronics manufacturing we did in India, originally added value of around \$14 billion dollars only,” explains Sanjeev Keskar, founder and CEO, Arvind Consultancy.

He adds, “Now imagine a market as big as \$160 billion and India getting only \$14 billion. It is a very, very alarming situation. What makes the situation grimmer is the fact that the market will evolve to be around \$360 billion dollars in the next five to six

The time is now

1. The world is looking towards a new destination for manufacturing electronics.
2. The government of India has introduced various PLI schemes promising billions of rupees as incentives for manufacturing electronics locally.
3. Global champions setting shop in India will need a lot of support from the local players.
4. India has almost become the biggest market for consumer electronics and smartphones.
5. Talent from India is already helping global companies create global products.

Where are the opportunities?

1. The lowest hanging fruit is IP and technology creation. Indians have already done wonders here, and it is time Indians start doing the same for India.
2. Start-ups can also start working in the field of ATMP. This is where India is lagging behind as compared to countries like Thailand, Singapore, and China.
3. Designing SoCs, or ICs, or ASICs is also a low hanging fruit. Indians working in multinational companies designing these prove that the talent pool does exist in India.
4. Semiconductors are a big opportunity, but these require a lot of investment. Start-ups can work on establishing a fabless semiconductor model in India like the one in Silicon Valley.

years. If we continue with low-value addition manufacturing, then the number of components and products we import, and the import bill, will keep on getting towards the increasing side.”

As a matter of fact, India imported 115,000 crore (1.15 trillion) rupees worth of electronics components during 2019-20, and 37 percent of these were imported from China. The bill of smartphone components that the country imported during this time was 56,039 crore (560.39 billion) rupees; 25,441 crore (254.41 billion) rupees worth of such components were imported from China.

It is worth mentioning here that India is the second biggest market for smartphones in the world. In absolute numbers, the country, during fiscal 2020, imported electronics worth 3.7 trillion rupees, according to Statista. The value of electronic products exported from India amounted to 829 billion (0.829 trillion) Indian rupees during the same fiscal.

“Almost 40 percent of an electronic product’s value is made up of the components used in it. These components include semiconductors, PCBs, capacitors, passive components, and a lot of other things. The problem is that 90 percent of these components are imported into India. It also means that the 40 percent market will be difficult to capture in the short term,” explains Keskar.

He adds, “The next 40 percent value comes from IP and technology. This is the area where India has great talent and is doing great work, but the problem is that this talent is either working for a global design centre or in a service company. The number of companies doing innovation as well as manufacturing are just a few! Even smartphones, one of the biggest selling verticals of electronics, do not get designed in India.”

Most of the smartphone manufacturing units in the country import CKD units and assemble them in India before selling them to Indian or international consumers. It is also to be noted here that despite 24 semiconductor companies having their large captive design centres in India, a fabless semiconductor ecosystem, similar to the one that exists in Silicon Valley, is absent in the country.

A report published by NEC and Assocham in 2018 notes that, while electronics manufacturing has been continuously growing since the past few years, the exports have remained constant at around \$6 billion in contrast to the imports that have been growing year on year.

“Why can’t we create IPs and technologies that create that 40 percent value. The last 20 percent value is into EMS, packaging, testing, quality, and all other aspects,” adds Keskar.

A lot, rather most of the EMS companies in the country, focus on setting